Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/02/2006
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Project Title:	Coordination of Turning and Reaching		
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Zip Code:	02454-9110	**Congressional District:**	7
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	DiZio, Paul (Brandeis University)		
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Task Description:	We have developed a turn and reach paradigm to assess post-flight disturbances of arm-trunk coordination. Our recent ground-based work has shown that simultaneous turn and reach movements involve high rotational velocities of the trunk that generate large, potentially disrupting Coriolis forces on the reaching arm and on balance. Nevertheless, reaching accuracy is preserved because the central nervous system programs compensation for the self-generated Coriolis forces. Failure to predict accurately Coriolis force magnitude produces disruption of coordination and potentially loss of balance. Our aims are: Aim 1-Parametrically evaluate Coriolis force influences on kinematics and dynamics of voluntary turning and reaching to targets under conditions of normal and increased body weight. Aim 2-Measure reaching movements and postural control after motor adaptation to rearrangement of the relationship between voluntary torso rotation relative to the feet and inertial torso rotation in space. Aim 3-Assess whether motor adaptation to Coriolis force perturbed reaching acquired in a rotating room transfers to turn and reach movements in a natural environment. These experiments will be of value in pinpointing factors contributing to post-flight postural and arm-trunk coordination		

	deficits. They will also have significance for understanding potentially contributing factors in populations at risk for falling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Coriolis forces on the arm produced by torso rotation during everyday activities are much larger than interaction torques between arm segments. Several patient populations, including stroke patients and Parkinson's patients have been shown to suffer from specific deficits in control of interaction torques among limb segments. The large Coriolis torques on the limb due to torso rotation may perturb both the limb and overall posture. Other work in our laboratory is investigating the possibility that labyrinthine-loss patients lacking semicircular canal inputs may have impaired coordination of trunk and limbs during voluntary turning and pointing movements.
Task Progress:	This report covers the second year of this three year project. A previous report gave details of Year 1 progress, in which two major experiments were completed: 1. We studied the coordination of turning and reaching movements in the 0g phase of parabolic flight. In 0g, arm and torso movement displacements showed large overshoots with subsequent rebounds relative to preflight amplitudes, despite having lower peak velocities. Transient derangements of turn and reach movements were seen when subjects were tested in 1 g, post-flight. These findings provide clear evidence that exposure to weightlessness immediately alters the coordination of turning and reaching and gradually degrades compensation for torso generated torques on the arm, which is manifest as postural and reaching errors post-flight. This establishes a basic mechanism sought by TRQ 14a and indicates where effort should be focused for development of countermeasures. 2. We completed an experiment investigating whether feet-to-torso "pointing" movements would adapt to movement-contingent inertial perturbations produced by a rotatable floor. During exposure to ipsi- or contraversive platform rotations that incrementally grew to 50% of the velocity of their torso rotation, torso rotation kinematics relative to the floor (as well as the targets and feet) stayed constant, although subjects were unaware of the altered motion of their body in space imposed by the floor and did not perceive their motor compensations. Upon return to stationary conditions, the initial torso rotation movements showed negative aftereffects with large and faster turns or smaller and slower turns following ipsi- or contradirectional floor rotations respectively. This experiment provide a new theoretical understanding of torso rotation and it provides the necessary condition for completing Aim 2 of our project, studying recalibration of natural turn and reach movements. These findings provide information about the basic mechanism of calibration of the turning portion of turn and reach movements, which is a partial answer to TRQ14a In the Year 2 of this project, we completed one new experiments, which completes Aim 2. We have also begun collecting data for the experiment designed to answer the question of Aim 3. Experiment 1. Introduction. Our first experiment of Year 1 we showed that exposure to 0 g conditions alters the motor control of torso motor control. This validates the turn and reach paradigm as a tool for studying adaptation to weightlessness. In the second experiment of Year 1 we developed a simple, lab-based method for rapidly generating strong adaptation of torso rotation. This provided a tool for studying a critical aspect of spaceflight adaptation in the laboratory, and it enabled us to complete Aim 2 of our project, which is to measure reaching movements and postural control after motor adaptation to rearrangement of the relationship between voluntary torso rotation relative to the feet and inertial torso rotation in space. Methods. Twelve subjects made turn and reach movements before and after undergoing incremental adaptation of torso rotation (without any arm movements). All subjects underwent adaptation of torso rotation to ipsiversive platform rotations, which results in an increased amplitude of torso rotation upon return to normal, stationary platform conditions. Results. The increased torso rotation speed relative to the intended speed generated larger Coriolis forces on the arm during post-adaptation turn and reach movements than duringpre-adaptation turn and reach movements. Since subjects were turning leftward, the Coriolis force on the arm was rightward. In this circumstance subjects showed rightward deviations of their arm relative to their torso. The turn and reach movements in the pre-rotation period also generated rightward Coriolis forces, but these forces were compensated perfectly and there were no reaching errors, because the torso component of the movement was executed as intended. Readaptation to stationary conditions restored normal torso-ram coordination, with a quasiexponential time course. Discusssion. This experiment proves that compensation for torso-generated forces on the arm is a feedforward process. Errors in execution of torso rotation thereby generate erros in arm movements. We are currently analyzed data on postural stability from this experiment Relation to Bioastronautics Roadmap. This paradigm shows provides a model for tracking resumption of normal coordination of body and arm movements in a behavior that is known to be altered in weightlessness. Futue experiments will attempt to utilized the floor rotation paradigm to increase the sensitivity of post-flight measures of postural re-adaptation. Experiment 2. Introduction. Aim 3 of this project is to assess transfer of motor adaptation to Coriolis force perturbed reaching acquired in a rotating room to turn and reach movements in a natural environment. Motor adaptation to Coriolis force perturbations in the slow rotating room takes place in a context where there is no sensory information available to indicate that the body is turning. During rotation, subjects always feel stationary but their initial reaching movements are deviated by Coriolis forces. They continue to feel stationary as they adapt to the Coriolis forces and regain straight and accurate reaches. It is possible that the new motor pattern learned is contingent on the registered state of body rotation, i.e. the feeling of being stationary, and will be fully expressed only in the same context. When rotation of the room stops and the subject has been at rest for a minute the sensory cues still define a "no rotation" context. This means there should be aftereffects in pointing movements made to targets not involving trunk rotation because the learned motor compensation will be executed despite the absence of trunk-based Coriolis forces. However, this hypothesis predicts that a post-adaptation turn and reach movement as opposed to a simple reach may be performed more accurately because self-rotation will be registered and the adaptation acquired in the rotating room is specific to a non-rotating context. Alternatively, if the

	adaptation acquired in the rotating room generalizes to reaching movements in different rotation contexts, then both a simple reach and a combined turn and reach movement each should show negative aftereffects following adaptation in the rotating room. Methods. We plan to have twenty subjects complete this experiment, and so far we have tests 6. The turn and reach table is mounted in the rotating room so that the start button is on the axis of room rotation. Each subject performs turn and reach movements before and after a period of rotation in which they perform simple reaching movements with no voluntary torso rotation. Results and Discusssion. The per-rotation part of the experiment is like our past reaching experiments in the rotating room, except that subjects will be standing instead of sitting. We will determine the endpoints and curvatures of reaching movements, and expect to find a pattern of deviations similar what we have seen previously. This will confirm that there was initially rightward deviation and ultimately adaptation to the Coriolis force perturbations. We also expect post-rotation simple reach movements which do not involve torso rotation to show endpoint and curvature deviations mirror image to the initial perturbations caused by Coriolis forces during constant velocity rotation. The critical comparison is whether the pre- and postadaptation turn and reach movements differ. We will determine the relative timing of arm and torso peak velocities as well as the endpoints and curvatures of hand path in external and joint space. If the internal motor compensation learned during constant velocity rotation in the slow rotation room is specific to the sensed state of self-motion (no rotation), then there should be no difference between pre- and post-rotation turn and reach movements. If the adaptation is more general, then it should result in aftereffects during turn and reach movements. The adaptive “compensation” learned in the rotating room would then combine with the usual compensation for self-generated interaction torques, and the arm would deviate left of baseline. Relation to Bioastronautics Roadmap. If the more general form of adaptation is found, then it means a regime of practicing turn and reach movements in 0g conditions may be helpful for preventing post-flight disturbances.
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